Dragon's Den

The catchy name of my Greenfoot game is "Dragon's Den". During the course of the game the player will control a character (an actor with a human sprite) with the goal of avoiding being hit by a fireball that the dragon (an actor using the snake sprite) shoots at them. In addition to the dragon's fireballs the player will also avoid being hit by monsters (actors with the ant sprite) with increasing speed. When the game starts one of the 10 monsters positioned at the top of the world will move in the downward direction with a delay until the next monster moves downwards. If the player makes contact with the monster the game over condition is met and they lose. Each time the player successfully avoids an ant and it touches the bottom of the world the player's score is increased by 1 and the ant is returned to its starting position. As the game goes on the delay between each ant moving downward is decreased, making the game more difficult as time continues. In addition to the monsters that the player must avoid, the dragon actor will also be moving back and forth horizontally at the top of the world while randomly creating a fireball actor at its position that will move downward at a faster pace than the ants and which will also have a larger hit box. The game is won when the player's score reaches 100, though this score may change later once testing is done to determine and adjust difficulty.

Player Actor: The player actor will be represented by a human sprite. The player actor will start at the bottom center of the screen when the game starts. The player actor will have methods for moving in both the x and y dimension. The player will have 3 game over conditions: if they are touched by a fireball, if they are touched by a monster, or if they touch the top border of the world where the monsters are located. These conditions will end the game and are checked by determining if the player and the monsters, fireball, or top border occupy the same space.. Within the game the player actor will be able to traverse from the far right side to the far left side of the screen and vice versa as though it is in a loop (like how Pacman can go from each side of the maze to the other through the tunnels at the ends).

Dragon Actor: The dragon actor will be represented by a snake sprite. The dragon will start at the very top center of the world when the game begins. The dragon will then move to the right horizontally until it reaches the far right boundary and begins to move left until it hits the leftmost boundary, switches directions, and the process repeats. During the course of the game a method will be used to randomly create a fireball actor at the dragon's position.

Fireball Actor: The fireball actor will be represented by a yellow ball sprite. It will move downwards at a faster rate than the monster actors and if it comes into contact with the player actor the game ends. Once the actor touches the bottom border of the world it is deleted and the score increases by 2.

Monster Actor: There will be ten monster actors at the top of the screen beneath the dragon. Using a getRandom function in the actor's class a number 0-9 inclusive will be generated. This random number will then be input into a switch operator which will call a monster method and tell one of the ten monster actors to commence their action. Once this occurs the selected monster will move downwards and if it collides with the player the game is over. If the player successfully dodges and the monster hits the bottom border of the world it is

moved to its starting position and the score increases by 1. There will then be a delay before the random and switch methods are called, this delay will shorten as the game goes on, thus making the number of monsters at once increase thus increasing the challenge.

Score Actor: The score actor is a very simple actor that will display, in text, on the bottom of the world the current score, which starts at 0, increases by 1 when a monster touches the bottom border, and increases by 2 when a fireball touches the bottom border. When the score is greater than or equal to 100 a win message will be displayed instead and the game will end.

Concept Google Doc Link: Dragon's Den

Instructions Google Doc Link: Dragon's Den Instructions

Github Link: https://github.com/ArnoldCDC271939/CSC170